

Expected Production Input Sources

Churn Prediction Model — CRM Scope

1. Purpose of This Document

This document maps each group of model features to the system that is expected to provide that data in production. For the hackathon, all data used was synthetic. This mapping is meant as a starting point for the data engineering team to confirm exact tables, fields, and refresh schedules before the model is connected to real systems.

2. Feature Group to Source System Mapping

Feature Group	Features Involved	Expected Source System	Expected Refresh Frequency
Patient Profile	Age, Gender, Region, Insurance Type, Disease Condition, Baseline Risk	CRM / Patient Master System	Low (updates rarely)
Enrollment Behaviour	Num Programs Enrolled, Num Enrollments, Num Withdrawn, Enrollment Channel, Enrollment Reason	Program Enrollment System	Medium (updates on enrollment events)
Interaction History	Total Interactions, Pct Follow Up Required, Pct Resolved, Pct No Response, Pct Escalated, Num Financial Assist Interactions, Num Adherence Counseling	Patient Support / Case Management System (e.g., call centre or support ticketing logs)	High (updates with every interaction)
Eligibility	Num Programs Eligible, Pct Enrollment Eligible	Eligibility and Benefits Verification System	Medium (updates on eligibility checks)

3. Assumptions

- Source system names above are expected mappings based on the synthetic data structure, not confirmed production system names.
- Refresh frequency is an estimate based on how often each type of data typically changes, and should be validated with the data engineering team.
- If the production platform is a data platform such as Microsoft Fabric, each feature group above should map to a corresponding Fabric data source or pipeline before go-live.

4. Next Steps

1. Confirm exact source system and table names with the data engineering team for each feature group.
2. Confirm refresh frequency for each source, so the model always scores on reasonably fresh data.
3. Set up a data pipeline that produces the 20 features in this exact format before they reach the model.